Conditional Critical Gabor Bases Give M^p Counterexamples for Every $1 < p < 2$

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Status. Candidate full negative answer to Question 5.1 of arXiv:2408.16593, likely valid, for human review.

1 Source question

Yu asks whether, for $g, \gamma \in M^p(\mathbb{R})$ with $1 < p \le 2$, every critical Gabor Schauder frame reconstruction that converges in L^2 (with respect to some ordering) must converge unconditionally on M^q for $p \le q \le p'$. The paper records the known $p = 2$ obstruction of Heil–Powell and explicitly leaves the existence of a counterexample for every $1 < p < 2$ open [1, Question 5.1].

unconditional Schauder frame. Whenever $\{x_n^*\}_{n\in\mathbb{N}} \subseteq X^*$ is a sequence such that Equation (5.1) holds for all $x \in X$, then we call $\{x_n^*\}_{n\in\mathbb{N}}$ an *alternative dual* of $\{x_n\}_{n\in\mathbb{N}}$ and Equation (5.1) is called the *reconstruction formula* of x. If $\langle x_n, x_m^* \rangle = \delta_{mn}$ for all $m, n \in \mathbb{N}$ and the reconstruction formula is unique for every $x \in X$, then we call $\{x_n\}_{n\in\mathbb{N}}$ a *Schauder basis* for X. A sequence $\{x_n\}_{n\in\mathbb{N}}$ is said to be an *unconditional basis* for X if it is a Schauder basis for X and the associated reconstruction formula converges unconditionally for all $x \in X$.

Let $\mathcal{G}(g, \gamma, \alpha, \beta)$ denote the pair of Gabor systems $\big((M_{\beta n}T_{\alpha k}g)_{k,n\in\mathbb{Z}}, (M_{\beta n}T_{\alpha k}g)_{k,n\in\mathbb{Z}}\big)$. Now, using the language of Schauder frames, Theorem 2.5(b) means that we can not only extend the reconstruction formula associated with a Gabor Schauder frame $\mathcal{G}(g, \gamma, \alpha, \beta)$ from $L^2(\mathbb{R})$ to $M^p(\mathbb{R})$ but also endow the reconstruction formula with unconditional convergence in $M^p(\mathbb{R})$ for all $1 \le p < \infty$ if the window functions g and γ are chosen from $M^1(\mathbb{R})$. A follow-up question motivated by the extension of Theorem 2.5 is formulated as follows:

Question 5.1. Let $g, \gamma \in M^p(\mathbb{R})$ for some $1 < p \le 2$. Assume that $\mathcal{G}(g, \gamma, \alpha, \beta)$ is a Schauder frame for $L^2(\mathbb{R})$ (with respect to some ordering). Do we still have that

$$f = \sum_{k,n\in\mathbb{Z}} \langle f, M_{\beta n}T_{\alpha k}\gamma \rangle M_{\beta n}T_{\alpha k}g \tag{5.2}$$

with unconditional convergence of the series for all $f \in M^q(\mathbb{R})$ and $1 \le p \le q \le p' < \infty$?

Note that since $(M^p(\mathbb{R}))^* = M^{p'}(\mathbb{R})$, we should not expect that Equation (5.2) can be extended to $M^q(\mathbb{R})$ for some $q > p'$. Due to Theorem 3.5, we see that the answer to Question (5.1) is true for some family of functions for all $1 < p \le 2$, for example, $\mathcal{G}(\chi_{[0,1]}, \chi_{[0,1]}, 1, 1)$. However, it was shown by Heil and Powell in [28] that there exist $g, \gamma \in M^2(\mathbb{R})$ for which $\mathcal{G}(g, \gamma, 1, 1)$ is a Schauder basis for $L^2(\mathbb{R})$ but the associated reconstruction formula does not converge unconditionally for all $f \in L^2(\mathbb{R})$. We mention that it is still open whether there exists a counterexample to Question (5.1) for $1 < p < 2$. Specifically, for each $1 < p < 2$ do there exist $g, \gamma \in M^p(\mathbb{R})$ for which $\mathcal{G}(g, \gamma, 1, 1)$ is a Schauder frame for $L^2(\mathbb{R})$ but the associated reconstruction formula does not converge unconditionally for all $f \in M^q(\mathbb{R})$ for

Here $M^s = M^{s,s}$, $T_k f(t) = f(t-k)$, and $M_n f(t) = e^{2\pi i n t} f(t)$. We answer the question negatively already at $q = 2$.

2 Claimed contribution

Theorem 1. *For every $1 < p < 2$ there are real, nonnegative, compactly supported windows $g, \gamma \in M^p(\mathbb{R})$ with the following properties.*

1. *For each Heil–Powell admissible rectangular enumeration $\sigma \in \Lambda$, the pair*

$$\big((M_n T_k g)_{(k,n)\in\mathbb{Z}^2}, (M_n T_k \gamma)_{(k,n)\in\mathbb{Z}^2}\big),$$

ordered by σ, is a Schauder basis together with its biorthogonal dual, and hence a Schauder frame for $L^2(\mathbb{R})$.

2. *Its reconstruction does not converge unconditionally for every $f \in L^2(\mathbb{R}) = M^2(\mathbb{R})$.*

Consequently Question 5.1 of [1] has a negative answer for every $1 < p < 2$ (take $q = 2 \in [p, p']$).

The construction has a little more quantitative content. If its endpoint parameter is $0 < \delta < 1/2$, then, for $1 < r \le 2$,

$$\gamma \in M^r(\mathbb{R}) \iff r(1-\delta) > 1. \tag{1}$$

3 Proof intuition

At critical density a window supported on one unit interval has Zak transform equal to the window itself. Heil and Powell observed that a window which vanishes like t^δ at both endpoints has squared Zak transform in the Muckenhoupt class A_2. It therefore generates a Schauder basis for their rectangular orderings, although it cannot be a Riesz basis because its Zak transform approaches zero.

The unique dual window is the reciprocal $1/g$ on the unit interval. Thus the only missing issue in the 2024 question is whether that reciprocal has the requested modulation-space regularity. Its two endpoint singularities are $t^{-\delta}$ and $(1-t)^{-\delta}$. Their integer Fourier coefficients have the same positive leading contribution, rather than cancelling:

$$\widehat{\gamma}(n) \sim 2\Gamma(1-\delta)(2\pi|n|)^{\delta-1} \sin\left(\frac{\pi\delta}{2}\right).$$

The compact-support Fourier-coefficient characterization of M^r then gives the exact threshold (1). Choosing $\delta < 1 - 1/p$ puts both basis windows in M^p, while conditionality of the basis supplies the required failure at $q = 2$.

4 Construction and proof

Fix $1 < p < 2$ and choose

$$0 < \delta < 1 - \frac{1}{p}. \tag{2}$$

In particular $\delta < 1/2$. Choose a real, nonnegative function $g = g_\delta$ supported on $[0,1]$ such that

$$g(t) = t^\delta \qquad (0 \le t \le 1/4),$$
$$g(t) = (1-t)^\delta \qquad (3/4 \le t \le 1),$$

$g(1-t) = g(t)$, g is smooth on every compact subinterval of $(0,1)$, and g is bounded below by a positive constant on $[1/4, 3/4]$. Such a function is obtained by a symmetric smooth interpolation. Define

$$\gamma(t) = \begin{cases} 1/g(t), & 0 < t < 1, \\ 0, & t \notin (0,1). \end{cases} \tag{3}$$

The Schauder basis and its dual

This is precisely the family in Heil–Powell's Example 5.11. Their Theorem 5.10 and the calculation $|Zg|^2 = |g|^2 \in A_2(\mathbb{T}^2)$ show that $\mathcal{G}(g,1,1)$ is a Schauder basis of $L^2(\mathbb{R})$ for every enumeration $\sigma \in \Lambda$ [2, Theorem 5.10 and Example 5.11].

simple Zak transform characterization of lattice Gabor Schauder bases for $L^2(\mathbb{R})$. We then construct examples of Gabor Schauder bases that have interesting properties.

Note that while Z_g has a natural quasiperiodic extension from $[0,1)^2$ to $\mathbb{R}^2$, the definition of $\mathcal{A}_2$ weight only depends on the absolute value of the weight, and $|Zg|$ is 1-periodic in each variable.

Theorem 5.10: *Let $g \in L^2(\mathbb{R})$ be given. Then the following statements are equivalent.*

(a) *The Gabor system $\mathcal{G}(g,1,1)$ is a Schauder basis for $L^2(\mathbb{R})$ with respect to every enumeration $\sigma=\{(k_j,n_j)\}_{j=1}^{\infty} \in \Lambda$. Further, if C_σ is the basis constant associated to the enumeration σ, then $\sup_{\sigma\in\Lambda} C_\sigma<\infty$.*
(b) $|Zg|^2 \in \mathcal{A}_2(\mathbb{T}\times\mathbb{T})$.

Proof (a)$\Rightarrow$(b): Suppose that statement (a) holds. Then by Lemma 5.1 we have that $1/Zg \in L^2(\mathbb{T}\times\mathbb{T})$. Further, by hypothesis, $\sup_{N,\sigma}\|S_N^\sigma\| \leqslant \sup_{\sigma\in\Lambda} C_\sigma < \infty$, so Corollary 5.9 implies that $|Zg|^2 \in \mathcal{A}_2(\mathbb{T}\times\mathbb{T})$.

(b)$\Rightarrow$(a): Assume $|Zg|^2 \in \mathcal{A}_2(\mathbb{T}\times\mathbb{T})$. Then, by definition of $\mathcal{A}_2(\mathbb{T}\times\mathbb{T})$, we have that $1/Zg \in L^2(\mathbb{T}\times\mathbb{T})$. Further, we have by Corollary 5.9 that $\sup_{N,\sigma}\|S_N^\sigma\|<\infty$. Lemma 5.1 therefore implies that $\mathcal{G}(g,1,1)$ is a Schauder basis with respect to each $\sigma \in \Lambda$. □

While Theorem 5.10 implies that if $|Zg|^2 \in \mathcal{A}_2(\mathbb{T}\times\mathbb{T})$ then $\mathcal{G}(g,1,1)$ is a Schauder basis with respect to every enumeration $\sigma \in \Lambda$, this does not imply that it is a Schauder basis with respect to every possible enumeration of $\mathbb{Z}\times\mathbb{Z}$, since that would imply that $\mathcal{G}(g,1,1)$ is a Riesz basis. In fact, by Theorem 2.8, the Riesz bases correspond to the subclass of weights in $\mathcal{A}_2(\mathbb{T}\times\mathbb{T})$ that are essentially bounded, i.e., bounded away from zero and infinity. The following example is inspired by Babenko's example from Ref. 60; cf. Ref. 36, Example 11.2, pp. 351–354, and gives examples of Gabor Schauder bases that are not Riesz bases.

Example 5.11. Fix $0<\alpha<1/2$. Assume that $g=g_\alpha \in L^2(\mathbb{R})$ satisfies the following:

(a) g is real-valued;
(b) $\mathrm{supp}(g)\subseteq[0,1]$;
(c) g is infinitely differentiable on every subinterval $(\delta,1-\delta)$, $0<\delta<1/2$;
(d) $g(t)=t^\alpha$ on $[0,1/4]$;
(e) $g(t)=(1-t)^\alpha$ on $[3/4,1]$;
(f) $g(t-1/2)$ is even;
(g) $g(t)\geqslant C>0$ for $1/4<t<3/4$.

Since g is supported in [0,1], a direct calculation shows that $|Zg(t,\xi)|^2=|g(t)|^2 \in \mathcal{A}_2(\mathbb{T}\times\mathbb{T})$. Therefore, Theorem 5.10 implies that $\mathcal{G}(g,1,1)$ is a Schauder basis for $L^2(\mathbb{R})$ with respect to every enumeration $\sigma \in \Lambda$. However, since $|Zg|$ is not bounded away from zero, Theorem 2.8 implies that $\mathcal{G}(g,1,1)$ is not a Riesz basis for $L^2(\mathbb{R})$.

We make some additional remarks concerning the windows constructed in Example 5.11. Note that since Zg is bounded, $\mathcal{G}(g,1,1)$ is a Bessel sequence, i.e., it has an upper frame bound. Since g is supported in [0,1], the dual window is easily seen to be the function that is $\tilde{g}=1/\bar{g}$ on [0,1] and zero elsewhere. The dual basis $\mathcal{G}(\tilde{g},1,1)$ possesses a lower frame bound but not an upper frame bound (compare Theorem 7.1).

Theorem 5.10 allows one to generate other interesting examples. Building on Example 5.11, the following example provides a Gabor Schauder basis that has neither an upper frame bound nor

Since $2\delta < 1$, both g and γ lie in L^2. The asserted biorthogonality can also be checked directly. If $k \neq \ell$, the interiors of the supports of $T_k g$ and $T_\ell \gamma$ are disjoint. If $k = \ell$, then $g(t-k)\gamma(t-k) = 1$

almost everywhere on $(k,k+1)$ and hence

$$\langle M_n T_k g, M_m T_k \gamma\rangle = \int_k^{k+1} e^{2\pi i(n-m)t}\,dt = \delta_{nm}.$$

Thus (3) is the unique biorthogonal window, exactly as noted by Heil–Powell.

They also prove $g \in M^1(\mathbb{R})$ [2, Theorem 6.1]; in particular $g \in M^p(\mathbb{R})$.

The reciprocal belongs to M^p

We record the endpoint calculation separately.

Lemma 1. *For γ in* (3)*, as $n \to +\infty$,*

$$\widehat{\gamma}(n) = C_\delta n^{\delta-1} + O(n^{-1}), \qquad C_\delta = 2\Gamma(1-\delta)(2\pi)^{\delta-1}\sin\left(\frac{\pi\delta}{2}\right) > 0. \tag{4}$$

The coefficients at negative integers are their complex conjugates. Consequently $(\widehat{\gamma}(n))_{n\in\mathbb{Z}} \in \ell^r$ if and only if $r(1-\delta) > 1$.

Proof. Take smooth endpoint cutoffs χ_0, χ_1 with $\chi_0 = 1$ near 0, $\chi_1(t) = \chi_0(1-t)$ near 1, and disjoint supports. By the exact endpoint form of g,

$$\gamma(t) = \chi_0(t)t^{-\delta} + \chi_1(t)(1-t)^{-\delta} + h(t), \qquad h \in C_c^\infty(0,1).$$

The standard endpoint oscillatory-integral formula (obtainable by the scaling $u = 2\pi n t$ followed by one integration by parts away from 0) gives

$$\int_0^1 \chi_0(t)t^{-\delta}e^{-2\pi i n t}\,dt = \Gamma(1-\delta)(2\pi n)^{\delta-1}e^{-i\pi(1-\delta)/2} + O(n^{-1}),$$
$$\int_0^1 \chi_1(t)(1-t)^{-\delta}e^{-2\pi i n t}\,dt = \Gamma(1-\delta)(2\pi n)^{\delta-1}e^{i\pi(1-\delta)/2} + O(n^{-1}).$$

In the second line the integer identity $e^{-2\pi i n} = 1$ has been used. The Fourier transform of h is rapidly decreasing. Adding the two leading terms and using $\cos(\pi(1-\delta)/2) = \sin(\pi\delta/2)$ proves (4). Because $C_\delta > 0$ and $n^{-1} = o(n^{\delta-1})$, the claimed ℓ^r criterion follows by the p-series test. □

Corollary 4.1 of [1], applied with $\alpha = \beta = 1$, states that for $1 < r \le 2$ the M^r norm is equivalent to the ℓ^r norm of the sampled Fourier coefficients of the restrictions to the unit cells. For clarity, the converse membership implication can be read without presupposing that $\gamma \in M^r$: take the ordinary Fourier partial sums

$$\gamma_N(t) = \mathbf{1}_{[0,1]}(t)\sum_{|n|\le N}\widehat{\gamma}(n)e^{2\pi i n t}.$$

Each γ_N lies in M^r, and Corollary 4.1 makes (γ_N) Cauchy in M^r exactly when $(\widehat{\gamma}(n)) \in \ell^r$. Its M^r limit agrees with its L^2 limit γ because $M^r \hookrightarrow M^2 = L^2$ for $r \le 2$. The reverse implication follows by applying the same norm equivalence directly. Lemma 1 therefore proves (1). The choice (2) gives $p(1-\delta) > 1$, and hence $\gamma \in M^p(\mathbb{R})$.

Failure of unconditional convergence

Heil–Powell also show that this Schauder basis is not a Riesz basis: its Zak transform is not bounded away from zero. In a Hilbert space, an unconditional Schauder basis is exactly a Riesz basis. Therefore, for any fixed admissible enumeration $\sigma \in \Lambda$, there is an $f \in L^2(\mathbb{R})$ whose expansion in this basis is not unconditionally convergent. Since

$$L^2(\mathbb{R}) = M^2(\mathbb{R}), \qquad p < 2 < p',$$

the exponent $q = 2$ is among those demanded in Question 5.1. This completes the proof of Theorem 1.

5 Verification and scope

The independent checks are recorded in `verification.md`. The accompanying script numerically checks the sign and scale of the endpoint constant, but computation is not used in the proof.

This result answers the Schauder-frame Question 5.1. It does *not* answer the separate question of whether the canonical dual of a genuine redundant Gabor frame inherits a given modulation-space regularity: the system here is a conditional Schauder basis and is not a Hilbert frame.

6 Provenance and novelty check

The conditional basis, its reciprocal dual, the A_2 argument, its failure to be Riesz, and $g \in M^1$ are all due to Heil and Powell (2006). The added observation is the sharp Fourier-coefficient audit of $1/g$, which puts the *dual* in M^p and thereby turns their construction into a counterexample to the explicit 2024 question for every $1 < p < 2$.

A bounded search on 11 August 2026 covered the run's registry, solution, attempt, and proof-gap indexes; exact-title and arXiv searches for arXiv:2408.16593; searches combining "Question 5.1", "Gabor Schauder basis", "modulation space", and "Heil Powell"; and later related Gabor unconditional-basis work. The 2024 source has only its original arXiv version, and no later paper found in that search states this answer. This supports, but cannot certify, novelty.

References

[1] P.-T. Yu, *Gabor frames with atoms in $M^q(\mathbb{R})$ but not in $M^p(\mathbb{R})$ for any $1 \le p < q \le 2$*, arXiv:2408.16593 (2024), `https://arxiv.org/abs/2408.16593`.

[2] C. Heil and A. M. Powell, *Gabor Schauder bases and the Balian–Low theorem*, J. Math. Phys. **47** (2006), 113506.